

Screen Sketches

4_SH_2: Pocket Deck

Austin Donald, Raine McKeller,
Mack Quinn, Matthew Ritland

Actors

3+ actors, each with multiple non-trivial functionalities, including brief description

1. **Primary Player:** The one who initiates a game. Has the privileges to:
 - a. Creates and configures a multiplayer game with modes and rules.
 - b. Invites friends or generates join codes.
 - c. Create and manage their personal account. This includes changing their username, email, and password.
 - d. Play different game modes.
 - e. Interact with friends via messaging.
2. **Friend:** The user who joins the game session initiated by the primary player. Has the privileges to:
 - a. Join multiplayer games via invite, friends list, or join code.
 - b. Interact with the primary player via chat, and emotes.
 - c. Can play the game interactively with the player, affecting the primary player's game.
 - d. Can also add other friends to the game or to their friends list, *if* the multiplayer game is configured to allow that.
 - e. Create and manage their personal account. This includes changing their username, email, and password.
3. **Spectator/Viewer:** A user who joins an active game without being a player. Has more limited privilege than the primary player and friend such as:
 - a. Watch active games in real time between friends. *(should we do past games too, like game recordings to replay a game?)*
 - b. Optionally view cards that are hidden to primary players and friends in an *active* game, *if* the multiplayer game is configured to allow that. *(if it is a recording, then they can toggle visibility of the cards on/off based on the spectator's wants)*
 - c. See which friends are in an active game

Non-Functional Requirements

3+ Detailed NFR's

- Support a minimum of 4 active users. (should it be more? For ex. There are 4 players and 3 spectators, is this multiplayer?)
- The game should take no longer than 200 ms to receive another player's move and update game state accordingly.
- Should take no longer than 1 second for any UI element to respond to maintain a smooth user experience.
- Users should not be able to see other user's email or password, but *only* the usernames of other users.
- Password should be encrypted when it is in the database (functionality? DO we want it?)

Tables and Fields

Properly identified all necessary tables and fields.

1. User: Tracks information related to user account
 - UserID - Primary Key (int)
 - Username - unique String
 - Email - unique String
 - Password - String
 - **DateCreated - LocalDate (represents year/month/day of creation)**
 - CurrentStatus - String (such as "In-Game", "Online", "Offline")

2. Friends: Tracks relationships and friends between users
 - FriendsID - Primary Key (int)
 - RequesterID - Foreign key that points to User.UserID
 - RequestID - Foreign key that points to User.UserID
 - Status - String ("Accepted", "Pending", "**Blocked**")
 - **DateBefriended - LocalDate (represents year/month/day of creation)**

3. DeckTemplate: Represents different versions of decks
 - TemplateID - Primary Key (int)
 - TemplateName - String (such as "Standard 52", "Standard 52 + Jokers", "No Face Cards", etc)
 - TotalCards - int
 - CardDefinitions - String (JSON form maybe? Says exactly what cards should be in the deck)

4. GameRules: Stores configurations for the different game modes
 - RuleID - Primary Key (int)
 - GameName - String ("BlackJack", "Poker", "Go Fish"...)
 - MinPlayers - int
 - MaxPlayers - int
 - TurnTimeLimit - int how long a player has before their turn ends, resulting in a skip or auto move

5. GameSession: Tracks active/past game lobbies

- SessionID - Primary Key (int)
- HostID - Foreign Key that points to User.UserID
- RuleID - Foreign Key that points to GameRules.RuleID (which game is being played)
- TemplateID - Foreign Key that points to DeckTemplate.TemplateID (which deck is used)
- GameState - String ("In-Progress", "Waiting For Players", "Completed")
- TimeStarted - LocalDateTime (stores year, month, day, hour, min, sec)
- TimeEnded - LocalDateTime (stores year, month, day, hour, min, sec, *is useful for seeing how long a game was in game history*)

6. PlayerSession: A table that tracks users and the specific game sessions they are in

- PlayerSessionID - Primary Key (int)
- SessionID - Foreign Key that points to GameSession.SessionID
- PlayerID - Foreign Key that points to User.UserID
- Role - String ("Host", "Guest", "Spectator")
- Score - int (for example, keeps track of how many chips/points)

7. UserStatistics: tracks a user's win/loss record

- StatID - Primary Key (int)
- PlayerID - Foreign Key that points to User.UserID
- RuleID - Foreign Key that points to GameRules.RuleID (to link stats for specific game modes, like poker stats vs blackjack stats)
- Wins - int
- Losses - int
- Draws - int

Screen Flow Diagram

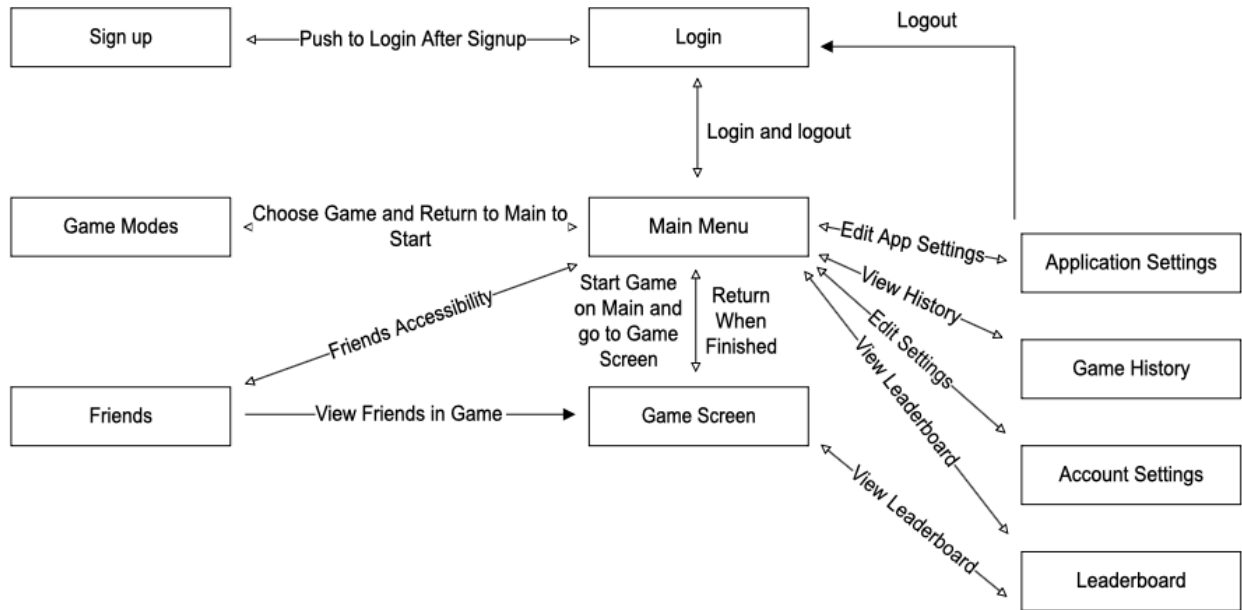


Figure 1: Screen Flow Diagram (Mack Quinn)

Blackjack Game Screen

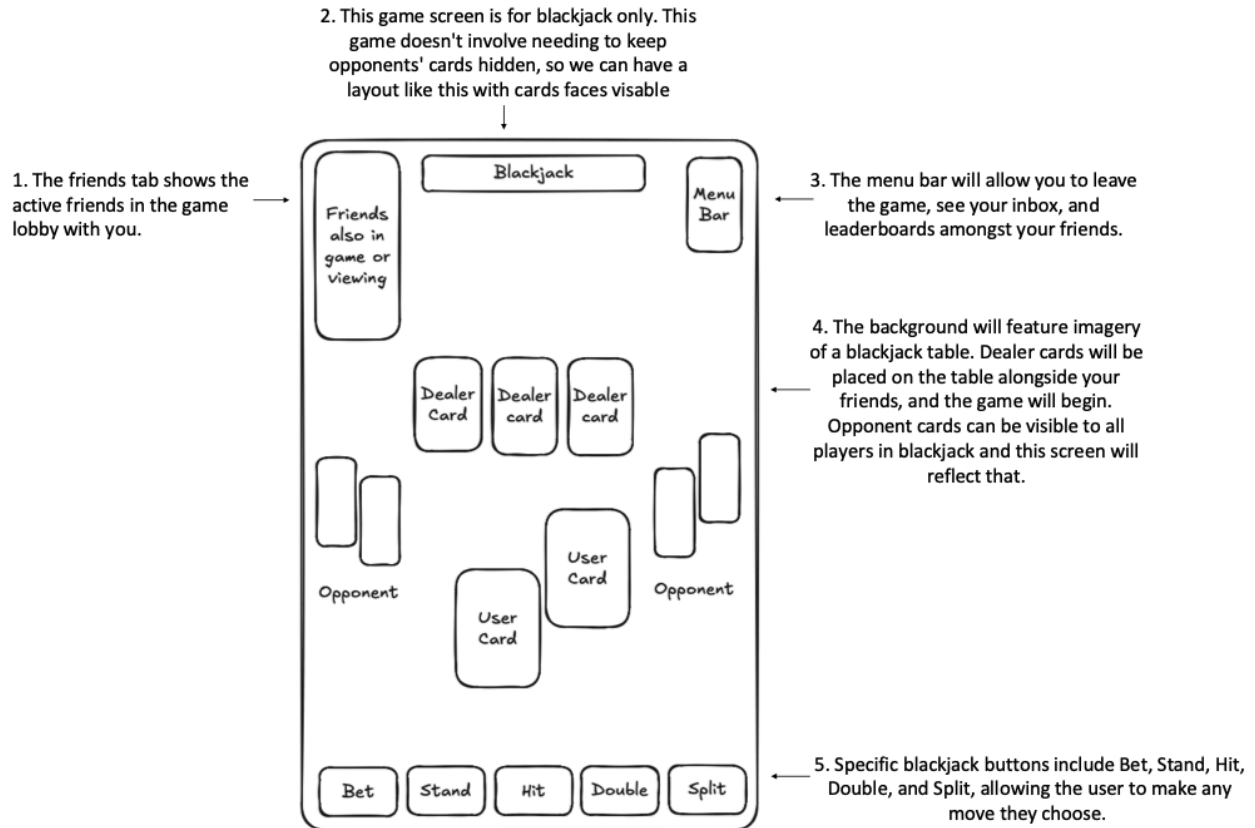


Figure 2: Gameplay Screen Blackjack (Mack Quinn)

This screen allows the user to play a game of blackjack online with friends. It has game specific details such as cards being visible to all playing, and the buttons on the bottom to control the game. Menus and buttons are situated on the edge of the screen to avoid clutter and allow the player to focus on the game. Menus should not change in between different games but rules and card layout may change.

This screen allows a player to play blackjack in a multi player environment with friends. It displays the dealers cards (4) along with the opponents cards (4) to simulate a real game. The top left corner (1) will show the friends in the game with the user. The drop down menu on the top right (3) will be used to allow the player to leave the game, access leaderboard for the game, and inbox. The control buttons on the bottom (5) will allow the player to control their moves and strategy.

Poker Game Screen

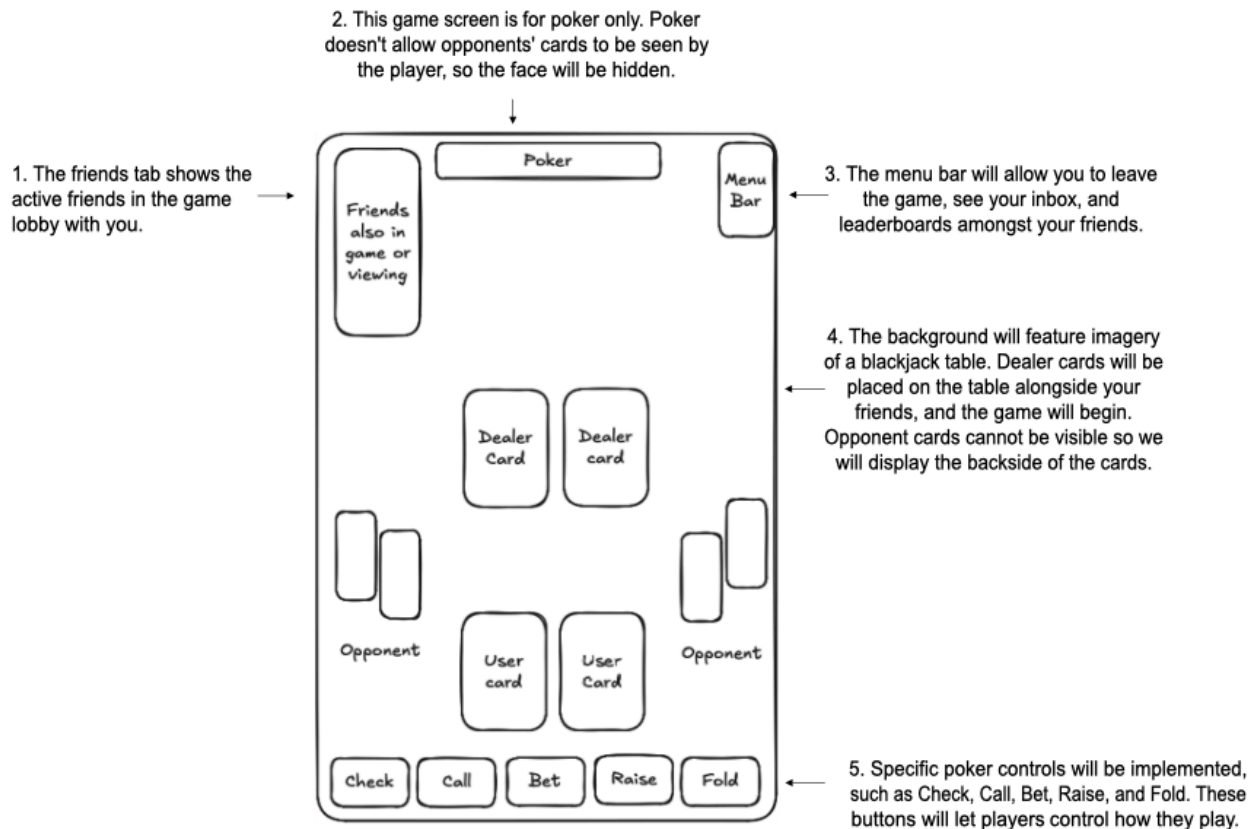


Figure 3: Gameplay Screen Poker (Mack Quinn)

This screen allows the user to play a game of poker online with friends. It has game specific details such as opponent cards being hidden to the primary player, and the buttons on the bottom to control the game. Menus and buttons are situated on the edge of the screen to avoid clutter and allow the player to focus on the game. Menus should not change in between different games but rules and card layout may change.

This screen allows a player to play poker in a multi player environment with friends. It displays the dealers cards (4) along with the backside of opponents cards (4) to simulate a real game. The top left corner (1) will show the friends in the game with the user. The drop down menu on the top right (3) will be used to allow the player to leave the game, access leaderboard for the game, and inbox. The control buttons on the bottom (5) will allow the player to control their moves and strategy.

Main Menu Screen

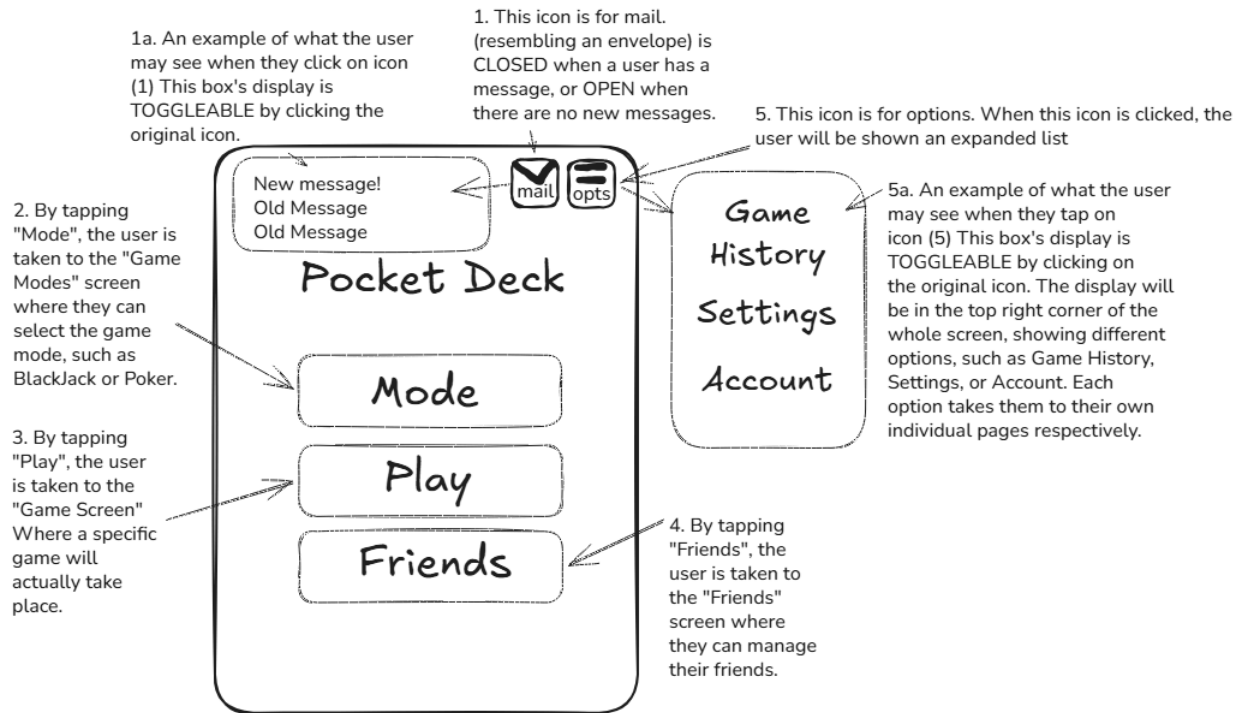


Figure 4: Main Menu Screen (Matthew Ritland)

This screen allows the user to navigate through the core features of the game. This page is structured so the user may easily identify and access (tap) the icons. With the expandable and collapsible icons, the user is able to access information only when they need it, so that the screen is not needlessly cluttered.

This screen lets the user, of whom is a primary player, view their mail from other players (1) within a collapsible message box (1a). They may also change enter the "Mode" screen through (2), enter the "Friends" screen to manage their friends through (4), Enter the game screen through (3), and lastly access less core features and their respective screens such as Game History, Settings, or Account through the menu icon (5) which contains the collapsible list where each word, such as "Game History", "Settings", and "Account" which take the user to that screen respectively (5a).

Game History Screen

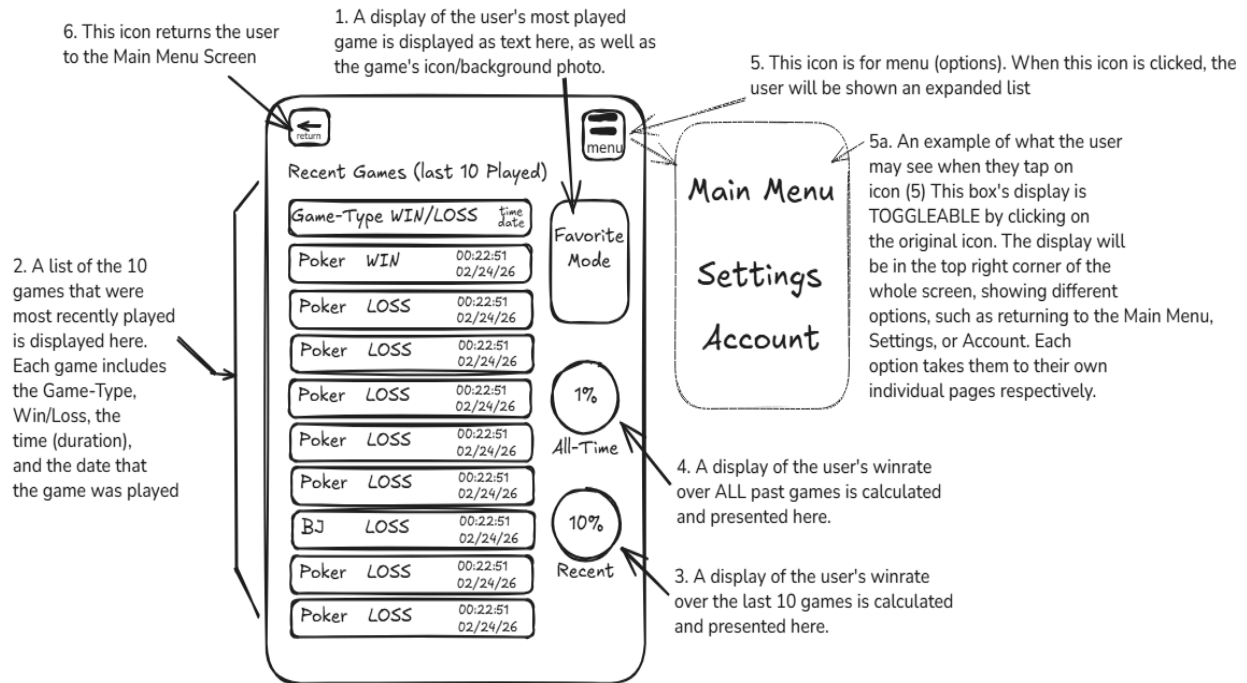


Figure 5: Game History Screen (Matthew Ritland)

This screen allows the user to observe their game history to observe their win rate and past wins and losses, which is limited to only the last 10 games played. It also displays their most played game mode within the box labeled "Favorite Mode." This page is structured so the user may easily identify and access (tap) the icons. With the expandable and collapsible icons, the user is able to access information only when they need it, so that the screen is not needlessly cluttered. They may also return to the Main Menu screen by clicking the box in the top left corner that has an arrow.

This screen lets the user, of whom is a primary player, view their 10 most recent wins and losses by looking at the displayed list (2.) They may also view their most played mode (1), recent win rate (3), and all time win rate (4). Similarly, they may access the menu (options) by clicking on the menu icon (5). This allows them to go to the Main Menu, Settings, or Account screens respectively (5a).

Screen Sketch #1

Exceptional effort and detail

Labeling is accurate and complete. All components are property numbered and named.

(insert below)

Title? (even though have it declared below in sample as well)

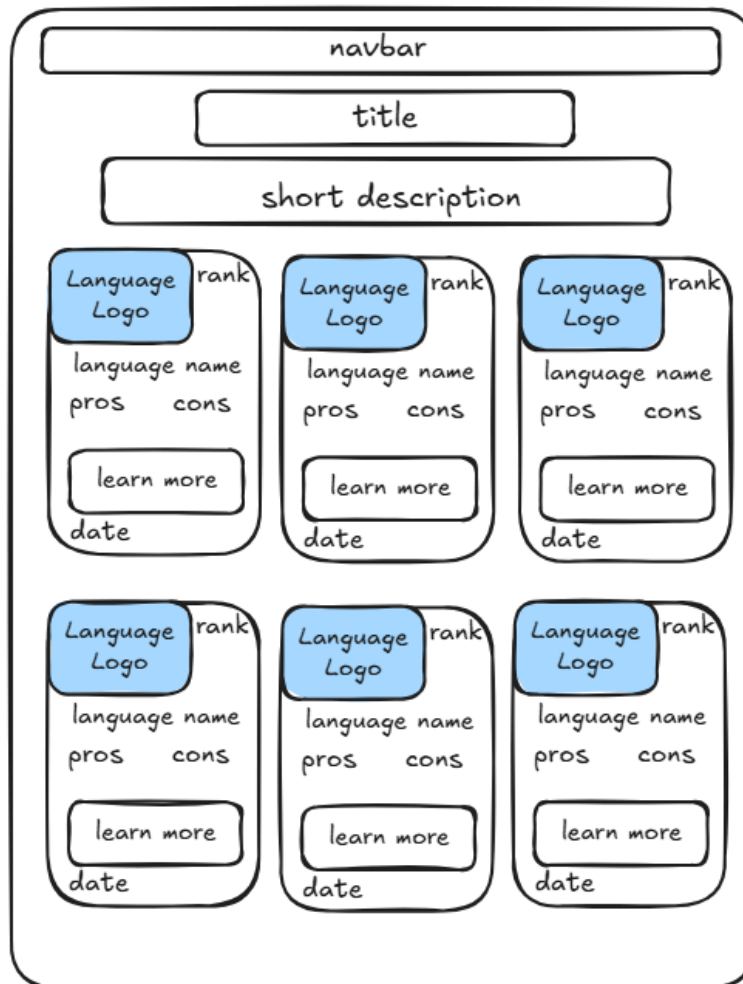


Figure 1: screen 1

Exceptional effort. Thorough descriptions of ALL components and events to reach screens.

Description description

Screen Sketch #1

Exceptional effort and detail

Labeling is accurate and complete. All components are property numbered and named.

(insert below)

Title? (even though have it declared below in sample as well)

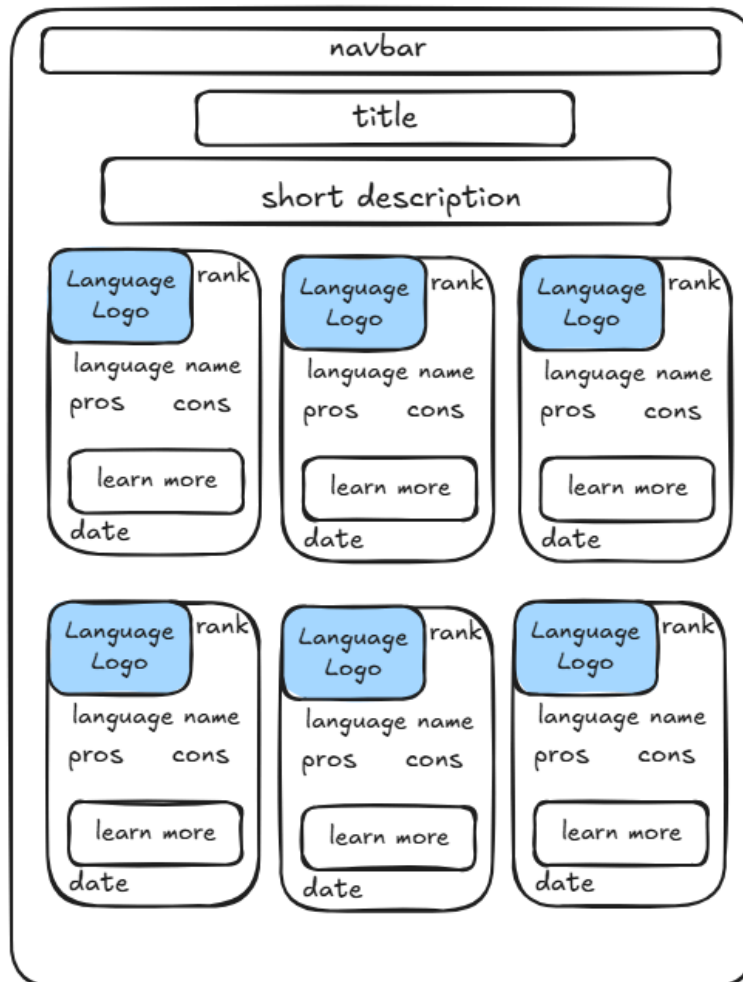


Figure 1: screen 1

Exceptional effort. Thorough descriptions of ALL components and events to reach screens.

Description description

Screen Sketch #1

Exceptional effort and detail

Labeling is accurate and complete. All components are property numbered and named.

(insert below)

Title? (even though have it declared below in sample as well)

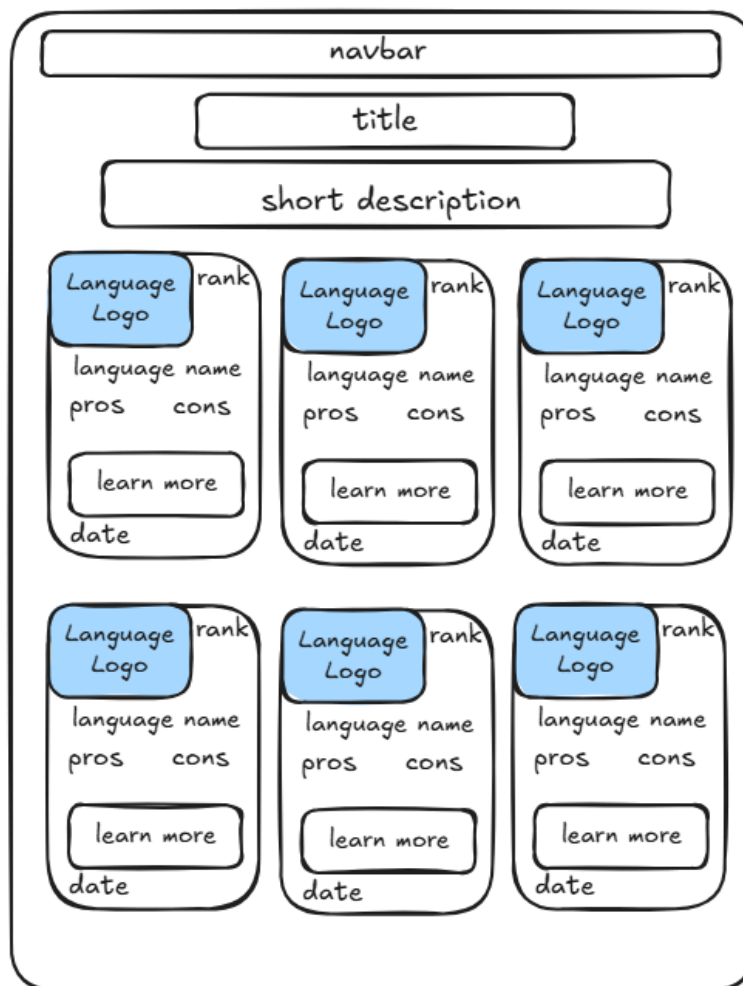


Figure 1: screen 1

Exceptional effort. Thorough descriptions of ALL components and events to reach screens.

Description description